

Consider a function $f(x, y)$ in two variables and a fixed point (a, b) . The two-variable Taylor series tells us that when the point (x, y) is close to (a, b) then we have

$$\begin{aligned} f(x, y) &\approx f(a, b) \\ &+ f_x(a, b)(x - a) + f_y(a, b)(y - b) \\ &+ \frac{1}{2} [f_{xx}(a, b)(x - a)^2 + 2f_{xy}(a, b)(x - a)(y - b) + f_{yy}(a, b)(y - b)^2] \\ &+ \frac{1}{6} [\text{third derivative stuff}] \\ &+ \dots \end{aligned}$$

If f has a local maximum or minimum at (a, b) then the tangent plane to the graph $z = f(x, y)$ at the point $(a, b, f(a, b))$ is horizontal, which means that $f_x(a, b) = 0$ and $f_y(a, b) = 0$. In summary, $\nabla f(a, b) = (0, 0)$. Conversely, any point (a, b) satisfying

$$\nabla f(a, b) = (0, 0)$$

is called a *critical point* of f . A critical point might be a local maximum, a local minimum, a saddle point, or something else. Near a critical point (a, b) , the behavior of $f(x, y)$ is governed by its 2×2 matrix of second partial derivatives evaluated at (a, b) :¹

$$f(x, y) \approx f(a, b) + \frac{1}{2} \begin{pmatrix} x - a & y - b \end{pmatrix} \begin{pmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{xy}(a, b) & f_{yy}(a, b) \end{pmatrix} \begin{pmatrix} x - a \\ y - b \end{pmatrix} + \dots$$

We will use the notation Hf for the *Hessian matrix*:

$$Hf = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}.$$

We also consider the *Hessian determinant*:

$$\det Hf = f_{xx}f_{yy} - f_{xy}^2.$$

Theorem (Second Derivative Test). Suppose that $\nabla f(a, b) = (0, 0)$.

- If $\det Hf(a, b) > 0$ then (a, b) is a local max or local min of f . To tell the difference, look at f_{xx} (or f_{yy}). If $f_{xx}(a, b) < 0$ then we have a local max and if $f_{xx}(a, b) > 0$ then we have a local min.
- If $\det Hf(a, b) < 0$ then (a, b) is a saddle point.
- If $\det Hf(a, b) = 0$ then we say that the critical point is degenerate. It could be a local max, local min, saddle point, or none of these.

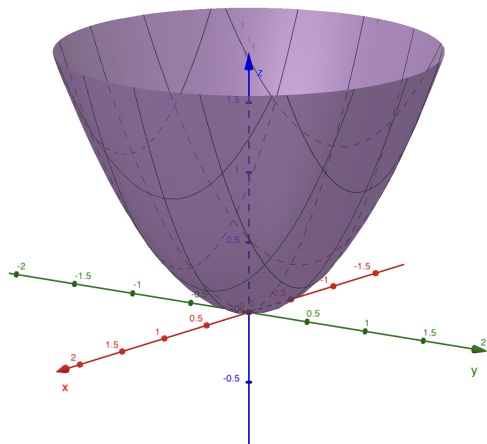
Here are the four most basic examples. In each case the Hessian matrix is constant, but that is only because these examples are so simple.

Local Min. Let $f(x, y) = x^2 + y^2$, so that $\nabla f(x, y) = (2x, 2y)$. If $\nabla f(x, y) = (0, 0)$ then $(x, y) = (0, 0)$ so there is only one critical point: the origin. The Hessian matrix is

$$Hf = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

¹Here we use some linear algebra notation. You can ignore it if you haven't taken MTH 210.

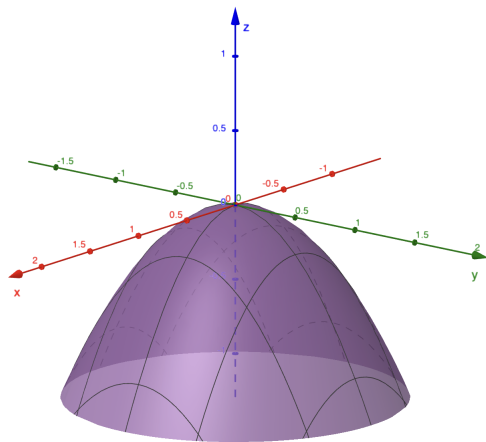
The determinant is $\det Hf = 4 > 0$ so the critical point is a local max or min. It is a local min because $f_{xx}(0,0) = 2 > 0$. Here is a picture:



Local Max. Let $f(x,y) = -x^2 - y^2$, so that $\nabla f(x,y) = (-2x, -2y)$. If $\nabla f(x,y) = (0,0)$ then $(x,y) = (0,0)$ so there is only one critical point: the origin. The Hessian matrix is

$$Hf = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}.$$

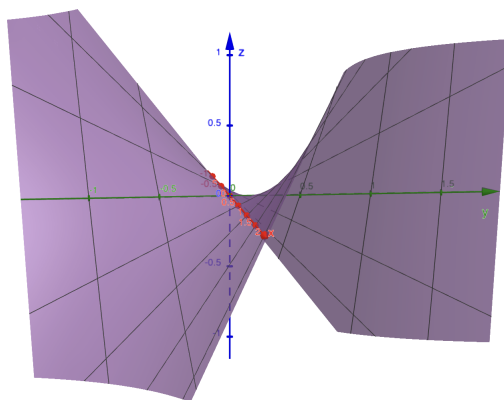
The determinant is $\det Hf = 4 > 0$ so the critical point is a local max or min. It is a local max because $f_{xx}(0,0) = -2 < 0$. Here is a picture:



Saddle. Let $f(x,y) = xy$, so that $\nabla f(x,y) = (y,x)$. If $\nabla f(x,y) = (0,0)$ then $(x,y) = (0,0)$ so there is only one critical point: the origin. The Hessian matrix is

$$Hf = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The determinant is $\det Hf = -1 < 0$ so the critical point is a saddle. Here is a picture:



Something Else. Let $f(x, y) = x^2$, so that $\nabla f(x, y) = (2x, 0)$. If $\nabla f(x, y) = (0, 0)$ then $x = 0$ so there are infinitely many critical points: $(0, y)$ for any y . The Hessian matrix is

$$Hf = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

The determinant is $\det Hf = 0$, so all of these critical points are degenerate. It turns out that they are all local minima, but the second derivative test doesn't tell us this. Here is a picture:

